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**r/dataengineering** • 1y ago
EbonyBlossom

What Path Did You Take to Become a Data Engineer?

Career

Hi everyone! I'm curious about the paths people took to become data engineers. Where did you start first? Did you build experience in another role before transitioning into data engineering, or did you aim for it right away?

For context, my current path focuses on learning SQL, systems analysis, operating systems, networking basics, scripting for automation, application support, and data visualization/reporting. I'm wondering if building experience in related roles (like data analysis or system administration) is the best approach before aiming for a data engineering position.

What helped you the most in your journey, and where do you recommend starting?



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↑ 93 ↓

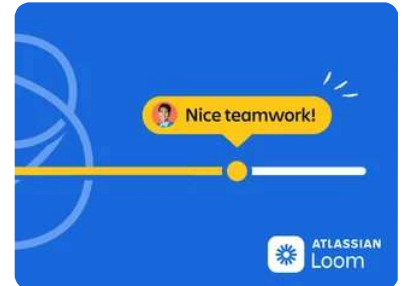
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MOD • 1y ago •



Stickied comment

**fauxmosexual** • 1y ago

My career started in the usual way that I think most of us will be familiar with: I joined a church repertory theatre company and one of the other actors offered me a job at his software company.

It was the 2000s when you could get a tech job just by looking kind of geeky.

↑ 87 ↓



[Skip to main content](#)[+](#) Create  2  34  ...**EbonyBlossom** OP • 1y ago

What kinds of tasks or projects as a data analyst helped you transition into data engineering?

  2  ...**Trick-Interaction396** • 1y ago

Someone needs an analysis from several different sources. Now I have to write query. Then schedule it. Then it's slow so I have to optimize it. Then it fails so I have to check the logs. Doing all these things requires understanding databases and engineering best practices.

 16  ...**itsthekumar** • 1y ago

Just curious how did you get into being a data analyst.

  2  ...**Trick-Interaction396** • 1y ago

Economics Degree

  6  ...**itsthekumar** • 1y ago

Gotcha. Cool.

 1  ...**Fun-Statement-8589** • 1y ago

Is it ok to ask? Currently, I'm learning... SQL, Python and inserting another learning that focused with Command Lines.

I took this free online course called Missing Semester by MIT that focuses in BASH (dunno if this is the right term)... i'm just wondering if I should delve deeper with BASH scripting or just focused on the command lines? I mean, making a function from vim or either nano with .sh is indeed a new language. Do I need to learn that language also?

Thank you.

 1  ...**eman_benz** • 1y ago

What is DE?

 1  ...

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[deleted] • 1y ago

Business analyst -> Business intelligence developer -> Data engineer.

I'd recommend practice SQL and Python medium/easy LC but outmost get a job that you can hop to DE later.

⊖ ↑ 59 ↓ ...



EbonyBlossom OP • 1y ago

Thank you!! I really appreciate it. I'm currently learning SQL the old fashioned way which is through textbook than buying courses from learn SQL and DataCamp just to save some coins I do pirate my books.

⊖ ↑ 11 ↓ ...



[deleted] • 1y ago

I'd recommend sqlzoo or anything that can get you on hand. For theory Google data engineering interview so you can check the topics I.e. ACID but they are just for interviews. It's difficult to understand without practice.

Market is bad now so get into anything that is remotely related should be good enough.

⊖ ↑ 7 ↓ ...



Busy_Pin6108 • 1y ago

Are you from the states?

⊖ ↑ 1 ↓ ...



[deleted] • 1y ago

Canada

↑ 1 ↓ ...



JBalloonist • 1y ago

This was pretty much mine too with some accounting (and degree) jobs prior to the analyst jobs.

⊖ ↑ 3 ↓ ...

[Skip to main content](#)[+](#) Create  ² 1  ... **day_paddler** • 1y ago

Great advice.

 1  ... **No-Map8612** • 1y ago **[deleted]** • 1y ago **homosapienhomodeus** • 1y ago

Data analyst to data science then data engineering - let me know if you want to learn more!

  10  ... **EbonyBlossom** **OP** • 1y ago

I want to learn more!!! Is it okay if I message you?

  2  ... **homosapienhomodeus** • 1y ago

no problem!

 1  ... **sealolscrub** • 1y ago

Did you like DE more or is it like you do more DE tasks compared to DS?

  2  ... **homosapienhomodeus** • 1y ago

I ended up preferring DE tasks because I learned more about all the components in the data engineering lifecycle and so I picked up new tools and more opportunities opened up for me in the role I was in. This obviously varies by your company, but regardless I found the DE tasks were more interesting.

 2  ... **itsthekumar** • 1y ago

Just curious how did you get into being a data analyst.

 **homosapienhomodeus** • 1y ago

I graduated with a physics degree, got interested in data and technology and several months later came across a data analyst role. Picked up some SQL and then grew from there!

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Can you elaborate on how and what exactly you self learned before starting the job. And what you learnt during the job. For data analyst, ds and de



homosapienhomodeus • 1y ago

Sure! Curiosity has been my starting point to be honest!

I did some SQL in one of my university modules and so that was the first thing I worked on before interviewing for data analyst roles. On the job, I was able to get more comfortable with advanced SQL as I had real use cases and large sets of data to play around with. I also learnt some dashboarding skills like Tableau and started working more with data engineers to understand how the data was being processed for analytical use cases.

This then led me to getting into Python and in 2020 data science was getting more popular, so I moved companies to start working on the Cloud and did some machine learning. I eventually picked more skills within the whole data engineering lifecycle and then moved roles to become a data engineer. I picked up DevOps skills for continuous integration and more.

I actually wrote an article on my journey some time back if you're interested in reading!
<https://moderndataengineer.substack.com/p/breaking-into-data-engineering-as>



InvincibleSolaire • 1y ago

Thanks a lot man



srodinger18 • 1y ago

So when graduated from uni, like the most of STEM students, originally I want to be a data scientist, so mostly I learned python and its data science library and a little bit of sql. However during university I also learn about HPC, linux, sysadmin and low level programming languages like fortran and c.

Then after months of job hunting, I got an offer as a data engineer in a consulting company. At first I just want to use this opportunity as stepping stone to become a data scientist, but after the time goes by, I started to become more comfortable as data Engineer, especially after saw data scientist colleagues mostly dealing with presentation and using ready to use models. Meanwhile in DE I learned many things not only from business perspective, but technical things as well



Naive-Risk9112 • 1y ago

I wanted to ask something ,is it okay if I message you

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Expense management, but make it do itself

[View More](#)[ramp.com](#)**Candid-Cup4159** • 1y ago

Why are you learning operating systems for data engineering? You rarely have to go that low level as a data engineer

**aacreams** • 1y ago

Operating systems knowledge has helped me a ton with the data platform development side of data engineering.

**EbonyBlossom** [OP](#) • 1y ago

I'm learning operating systems because I'm studying IT with a focus in Systems.

**Candid-Cup4159** • 1y ago

Ah, then by all means. No knowledge is wasted

**ChipsAhoy21** • 1y ago

Business analyst with a bit of VBA to automate some macros (mostly recording them and then tweaking the code)

Auditor who had to occasionally had to read code to make sure numbers weren't being manipulated in data pipelines, used sql to extract queries

Data analyst

Data Scientist

Data engineer

All over the course of about 10 years

[Skip to main content](#)[+](#) Create **mosqueteiro** • 1y ago • Edited 1y ago

Started from a Data Science bootcamp but I enjoyed making the tooling much more and whaddya know, most companies need data engineers, especially the ones who hire data scientists first 😄

Got a BS in Mechanical Engineering with (almost) minor in CS before the bootcamp so that helped a little too

What helped most was projects that I could point to and talk about like what it was, what it solved, some challenges faced and how they were overcome. Honestly, being able to talk about how you solve problems is huge. My experience is that Data Engineers kinda have to solve lots of different problems all over the place. Start with building strong SQL and programming skills. Those will be a foundation to build other skills off of.

**[deleted]** • 1y ago • Edited 1y ago

That's very encouraging to read! I'm in the process of completely change my career, and after taking few short college courses in coding and Python, I'm starting a boot camp in Data Engineering next week. I'm not sure how helpful it would be, especially considering the competition with uni graduates.

**mosqueteiro** • 1y ago

I, personally, don't think that boot camps are enough to get someone ready to be a Data Engineer. I think a 2 year course focused on Data Engineering is really needed. I don't know if any of those exist though and while the boot camp isn't enough, IMO, it's still going to be very helpful. There's few, if any, Data Engineering programs at the unis so competition will be more of having a degree vs not — relevant degree is more helpful (CS, Math, or Eng). Also, lots of companies don't even understand what a data engineer does or needs to be able to do. So with a boot camp, solid programming and SQL skills, and (your own) data engineering projects under your belt plus networking and grit you can likely find somewhere to start your career.

**[deleted]** • 1y ago

I agree with you on the fact that three months of boot camp isn't enough to secure a junior position, and my BA in graphic design won't be of any help. What did you do after the boot camp, if you don't mind me asking?

**mosqueteiro** • 1y ago

Actually, I think your BA in graphic design probably does help quite a bit even if it's just that you have a degree —although it's also a technical or technical-adjacent degree so more points in your favor. It's not that you can't get a job with just a boot camp, plenty of people do, it's just that Data Engineering is pretty vast in what it touches. This is especially true at small companies or companies with small or brand new data teams. If anything, search for any job in data when you get to the job search point (this should be part of the boot camp). There are a lot of people that start as a Data Analyst or Business Intelligence Analyst and move into

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work. If you are excited by the things you're learning you know you're on a good path. If it feels like a chore then maybe reflect on if you really like this or what the motivation was in the first place.

I got a job at a startup right out of my Data Science boot camp as a Data Engineer. My capstone project was a deployable and scalable data science environment using docker containers. I ended up making something similar at the startup. This company didn't even have a database when I started. Everything was in CSV and XLS files. So there was a lot I could do right off the bat. I basically have had a large lab to experiment and learn. I do wish there had been someone more senior that I could've learned from their experience directly and gotten guidance on what was important and how to structure things. I feel like I've re-learned, through trial and error, quite a bit that I might've gotten from working with a senior Data Engineer. I'm actually only finding more things to learn as time goes on 😄

Data engineering is still in its adolescence. Things are changing quickly and fundamentals are slowly starting to coalesce. It's a great time to be in data!



[deleted] • 1y ago

Thank you very much for sharing all these insights; they are incredibly helpful!



InterestingDegree888 • 1y ago

I was in Customer Service Help Desk in a call center and started teaching myself SQL and DBMS, started playing with some automation to create reports (dts, sql server, and crystal reports... it was a long time ago) then moved into a financial / center analyst role, decided to go off and become data consultant, went back to corporate as a analyst that was really doing DE and DA work but we didn't have those titles back then, finally into a Sr DE role and then eventually a director role ...

I have a video walking through what I think that the basic skills should be and what order you should pick them up in (Roadmap) as well as walking through a project based resume...

My advice - find any position that is remotely data related and apply for them... get some practical work experience and then move up into a DE role... Also, find a project where you are passionate about the dataset and do the project from inception (design / architecture), the engineering part (pipelines, modeling, cleaning, etc.) and the analysis / viz as well.

Here is the link to my video if you are interested:

<https://youtu.be/sj8IWYrFTBQ>



[deleted] • 1y ago

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engineering and didn't ever plan on it. Working at a startup as a SWE I was thrown into a lot of tools but having Databricks experience made a lot of companies interested. I never saw myself as a data engineer but those were the roles recruiters would box me in.

Tbh exposure to all ends of a pipeline just made learning data engineering easier and natural for me.



AchillesDev • 1y ago

Early career my hiring managers (after hiring me) pointed out my graduate neuroscience degree was a big selling point for hiring me.



LettuceElectronic995 • 1y ago

android developer -> data engineer



DerelictMan • 1y ago

Me too! High five. 🙌



LettuceElectronic995 • 1y ago

yeah, that rare breed.



TemperatureMoist7636 • 1y ago

Modern data engineering is almost cloud based. Azure and AWS are popular cloud providers where most companies rely on. SQL and Python is necessary for a data engineer in my opinion. If you stick to Azure, there is a good course on udemy by DataLearnHub to take you through Azure data engineering essentials.



FlyKillaDataGrl • 1y ago

Art major (lol) --> Data Science bootcamp --> Data Scientist --> Data Analyst --> Data Engineer



[deleted] • 1y ago

What boot camp did you attend?



[Skip to main content](#)[+](#) Create**[deleted]** • 1y ago

Thank you

**7tony_stark7** • 1y ago

Recently I have joined a company as a role of data engineer. In interview, they will put more emphasis on only three things for freshers:

1. SQL (You should have a great grasp on this)
2. Database (Concepts such as Normalization, Modeling etc.)
3. Logic for Programming like DSA (In any language but pref. in Python)

After three rounds of interview, I joined the company and in all rounds, most of the questions are from this three things.

**Paige_2002** • 1y ago

I wanted to be a data engineer but didnt shoot for it right away. I think it's important to gain some domain knowledge initially as an analyst/scientist before going that route - not necessary but helped me for sure, especially since my background is in math/stats.

I began work as a data analyst for a small-medium sized company. The previous data warehouse manager had left a few months before I joined. I was a pretty good SAS developer and had expressed to my boss engineering was the route I wanted to go down so she geared those tasks more for me. Eventually it became my main job.

I agree with just about everyone in the comments - get your foot in the door and take on DE tasks when/where you can :) I hope this helped.

**EbonyBlossom** [OP](#) • 1y ago

Thank you 🙏. That's exactly what I plan on doing I just needed second opinion

**gabbom_XCII** • 1y ago

Lead Data Engineer

FP&A / Excel Monkey -> Business Intelligence, noticed my alteryx/python pipelines were utterly shit and started (really) learning SQL and also big data tools like Hadoop/Hive/Spark and AWS, learned data lake/data warehousing concepts and landed a Data/Analytics Engineer role.

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I got a college degree in teaching woodshop. No lie.



SilentSturm • 1y ago

Digital marketing -> Data engineer.

Found out I like tech and data after spending years working with martech tools.



iammerelyhere • 1y ago

Panel beater>Customer Service Clerk>Accounts Receivable>DBA>Visual Basic Programmer>.Net Developer>Business Analyst>Data Engineering. Currently a Project Manager.



NaturalBornLucker • 1y ago

My path to DE job was somewhat odd. I've graduated (not USA) with a CS degree, worked a couple of years as SWE and QA, then got into ITSM\ITIL field as a technical consultant. It was fun and games until ten years later being a tech lead I've realized that I have almost nothing to learn, sick to death of the system and don't have an opportunity to grow or even switch to manager role. So I quit my job and thought about "what I did like in my job?". The answer was - integrations. So for a 4-5 months I studied modern frameworks and languages (even had to start SQL from the scratch and learn relational algebra) and then started to hunt jobs. I was lucky and landed good middle DE role in a bank. Then it became easier. Now, two years later, I'm interviewing for middle+/senior roles feeling comfortable enough and hoping in 2-3 more years to grow to a team/tech-lead again.



sujitpal_reddit • 1y ago

I graduated CS at a time when DS/DE weren't things, closest we learned about this stuff was Operations Research and Numerical Methods. I started life as a software customer support engineer at a computer manufacturing company, where we did mainly OS level installs and troubleshooting, then moved to their application software team (COBOL/C with proprietary DB), then got a job at an installation where they just purchased a UNIX based minicomputer but had no programming support, so I became the sysadmin / sysadmin / application developer. I then left to join a consulting company based on learning the DB system from my previous job, initially as application developer but then got a chance to transition to DBA when I did some work on perf improvement that needed an understanding of system tables. Following that I got certified and was hoping for a real DBA job but got a job as an "application" DBA where I built stored procedures as the data layer of a Java application. Soon after the Java devs started leaving for better jobs, and I had nothing to do (stored procedures were all done) so I transitioned to operations and bug-fixing for the Java stuff. I then got a backend Java job and they taught me web development. Following that I moved to

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learning NLP (and a little ML to support NLP). I also rebuilt the platform on Solr and helped re-write the indexing backend in Scala to make it faster, which got me my current position, where I started as big data engineer (Hadoop followed by Spark), then moving to ML (mainly Neural Networks) to support NLP, search and recommendation systems. My focus now is more ML engineering than data engineering, but I am part of the DE team so I guess that makes me a DE.



[deleted] • 1y ago

Developing Oracle Forms/Reports 6i -> CS degree -> ? -> DE



inrusswetrust12 • 1y ago

Is it a guessing game for the ?



aacreans • 1y ago • Edited 1y ago

Completely fell into it as a CS major. Got a data engineering internship offer while spam applying to SWE internships and ended up liking the field and being good at it so I just stuck with it.



AutoModerator App **MOD** • 1y ago

You can find a list of community-submitted learning resources here:

<https://dataengineering.wiki/Learning+Resources>

I am a bot, and this action was performed automatically. Please [contact the moderators of this subreddit](#) if you have any questions or concerns.



Pristine_War_3329 • 1y ago

Hello guys, there are awesome information about this post. I am also in the same place. Is there anyone who is in Germany for Data Engineer? Let's connect



ianitic • 1y ago

Studio production coordinator -> procurement specialist -> implementation consultant -> project coordinator -> tier 2 support -> process automation engineer -> data engineer

[Skip to main content](#)[+](#) Create**EbonyBlossom** OP • 1y ago

Wooww that's impressive!! Thanks for commenting I really appreciate it.

**ianitic** • 1y ago

Thanks haha!

Some of those titles may sound nonsensical on the path to the title but I always chose jobs where I had a lot of autonomy in how specifics were done which allowed me to do a lot of scripting and data processing.

Like the production coordinator and procurement specialist roles I was also got all the responsibilities of a business analyst rolled into it.

**ProcessOk6477** • 1y ago

Data analyst -> data scientist -> data engineer

**nightstlikethese29** • 1y ago

Data analyst -> data engineer

I had the title data analyst for a year and a half and I've had the title data engineer for a year. Realistically, I've been doing data engineering work for all but 6 months of my time in data.

**chrisgarzon19** • 1y ago

CEO of Data Engineer Academy

BIE is an underrated role to bridge analyst to engineer

**CasteliaLyon** • 1y ago

Business Analytics degree -> consulting doing DE ?

I'm curious to see why this isn't a more common path here in this thread.

**hauntingwarn** • 1y ago

[Skip to main content](#)[+](#) Create  ²**AchillesDev** • 1y ago

My first job out of school (non CS degree, dropped out of a science PhD program to become a coder because I liked it more) was a standard boring enterprise desktop application company. Nice people, severely underpaid, but it was my foot in the door. Dodged a layoff by going to another company as a software engineer. I walked in and saw their ETL pipeline (I didn't know what data engineering or ETL was at the time. This was...2015?) and knew I wanted to work on it. Got the job, was there for 2.5 years, started my own company, then moved across the country to work at startups where I was able to wear multiple hats and get more experience on the MLE/infra side of things and have been bopping around different subdisciplines related to it since then. Most lately computer vision (since...2019?) and LLMs.

**siddartha08** • 1y ago

4yrs Operations Analyst -> 3yrs Business Intelligence Analyst(Data Engineer job duties) -> Data Engineer.

**sib_n** • 1y ago

Senior Data Engineer

See related post:

https://reddit.com/r/dataengineering/comments/1i6xclc/how_many_of_you_are_selftaught_data_engineer/

**EbonyBlossom** [OP](#) • 1y ago

Thank you I will!!

**InterestingDegree888** • 1y ago

Customer Service Help Desk > Data Guy (because we were not called data engineers back then) > Data Consultant (still no one called us DE) > Data Analyst that did Analyst and DE (because still no DE) > Senior DE > Director of Data & Analytics...

Find your path, everyone's is different. Focus on SQL to start. Understand modeling, normalization, familiarize yourself with a few ETL / ELT tools, play with a cloud platform (AWS or Azure are the more popular corporate ones), then hit python and keep branching out from there as you work...

Books are great, but find a dataset you are passionate about and do a full end to end project. Something hitting a API would be awesome. but do the whole thing all the way through visualization to show you have all those skills.

Apply for anything remotely data related to start getting some practical work experience on your resume...

[Skip to main content](#) Create<https://youtu.be/sj8lWyRfTBQ>**ResidentAd138** • 1y ago

Physicist with parallel computing and HPC -> Data Engineer

**k00_x** • 1y ago

Enjoyed playing snes - hacked a save - learnt how the cartridge memory worked - carried on messing with windows pcs - studied Satellite technology at uni (mistake) - majored in operating systems - Interned for IT firm that migrated blockbuster video on to windows - got a job as a contract tech support for web hosting - exposed to pretty much every type of data - solved all kinds of data issues so my boss could get paid - fast tracked to product manager - built all BI for company - burnt out - said f*ck tech and became an apprentice patisserie - became coeliac, nearly died - became a healthcare procurement specialist analysing where to spend 1/2 a billion a year - got a good rep in healthcare data - worked in mental health services - brought an unused data source into SQL server, changed everything, moved from excel to dashboards - moved to emergency ambulance setting - built in a platform that can handle 'live' updates with almost no budget - spend all day looking at new jobs never apply.

**modern-pineapple** • 1y ago

Try <https://www.interviewquery.com/>

**idkwhatimdoing069** • 1y ago

Lvl 1 IT support -> Lvl 2 Support -> Poached by a client for knowing a touch of python. I'm here by accident but enjoying the ride

**james2441139** • 1y ago

Senior data architect here. I have a bachelors in engineering and masters in computer science. Engineering and CS degrees definitely helped me to learn the basics. And then I worked data engineer for about 4 years before moving up to be a senior architect. My best advice is learn with personal projects and focus of cloud platforms (AWS or Azure mostly). Feel free to dm me if any questions. Happy career path!



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-> «Business Developer». Just a title. I actually was a one-man dev team for small-scale business applications, mainly web apps on svelte and react, integrations and automation.

-> Consulting. Hired by a client as temporary Product Owner for their Data Platform while actual PO was on extended parental leave. Started off managing their migration from On-prem SSAS to Synapse. Talked them into giving me dev privileges. Currently on my 3rd year in this role, but now I'm 80% Data Engineer and 20% PO.

I have an MSc in Industrial Economics. Solid business understanding, data literacy and technical skills (SQL + Python) will take you a long way! I've also found that understanding accounting principles and terms can be very useful



sealolscrub • 1y ago

Reports developer > database administrator > data migration specialist > DE

DE isn't really a popular name 10 years ago, mostly it is <insert database tech name> developer. But since most of the companies switched to cloud and the data keeps on getting bigger. The need to migrate the dataplatform into cloud and into an analytics platform efficiently to help on business decisions. Thus made data scientists, analysts and engineers in demand.



Old-Cardiologist-334 • 1y ago

BS in Civil Engineering>5 years in Construction>Software Engineering Bootcamp>DE Internship>DE



dawndaydreams_ • 1y ago

I'm very aware I lucked into what is now a competitive industry, but for me, it was:

Standard Office Admin > Excel Analyst > Data Analyst > BI Developer > Data Engineer

Edited to say: I'm all completely self-taught, with no college/university education.



Anjalikumarsonkar • 1y ago

Many data engineers begin their careers as data analysts, software engineers, or system administrators before transitioning into data engineering. Key skills needed for this role include strong proficiency in SQL, Python, experience with ETL processes, and familiarity with cloud platforms like AWS or GCP. Your current path seems promising; gaining hands-on experience with databases and data pipelines will be beneficial as you work to enter the field of data engineering.



 Create



 [deleted] 1y ago